

# Stylometric Features and Evaluation

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Contribution II to the joint report with Kai S. Kurono

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## Abstract

This contribution complements the word-based Reuters exploration with explicit style features and a larger evaluation. It retains William Kazer, Tim Farrand and Patricia Commins, the same 147 reference articles, and the same three mystery texts. It additionally evaluates 150 articles from the official test partition. A TF-IDF nearest-neighbour model achieves 91.3% accuracy, a style-based nearest-neighbour model 69.3%, and a style-based author-centroid model 82.7%. These results suggest that both representations contain useful information, but they do not establish topic-independent authorship recognition.

## 1 Question and relation to the joint project

The word-based contribution by Kai S. Kurono shows how word-based representations organise articles by the selected authors. However, highly weighted features such as **china**, **million** and company names also describe subject matter. The complementary question is therefore whether simple, interpretable style features can support attribution on the same documents.

The investigation keeps the existing nearest-neighbour decision rule for a direct representational comparison. A NumPy author-centroid baseline tests whether an average profile is more useful than one individual reference article. PCA remains an exploratory visualisation and is not used as classifier input.

## 2 Data and evaluation design

The data come from Reuter\_50\_50 [1]. Each of the three authors has 50 articles in **C50train** and 50 in **C50test**. The existing notebook’s alphabetic filename ordering is retained: one article per author is reserved as a mystery text, leaving 49 reference articles each.

| Author           | Reference | Mystery | Official test |
|------------------|-----------|---------|---------------|
| William Kazer    | 49        | 1       | 50            |
| Tim Farrand      | 49        | 1       | 50            |
| Patricia Commins | 49        | 1       | 50            |
| Total            | 147       | 3       | 150           |

The reference set is identical for all three models. Vocabulary selection, inverse document frequencies, feature scaling, PCA and centroids use only these 147 articles. Neither the mystery texts nor

the official test articles contribute to fitting. The three reserved training articles are not added back for the larger evaluation, ensuring that its reference set remains the same as the word-based experiment.

Raw-file and stripped-text hashes were checked for exact matches between the reference set and each evaluation group. None were found. This does not exclude closely related or near-duplicate news stories.

### 3 Style-feature extraction

The feature extraction in `stylometry/features.py` preserves the measurements of the preceding local spaCy implementation. Line endings are normalised, HTML entities are decoded, and outer whitespace is removed. Capitalisation and punctuation are retained. The English model `en_core_web_sm` supplies tokenisation and sentence boundaries [2]. Alphabetic tokens form the word counts.

A compact, predefined set of 26 features is used:

| Group            | Count | Definition                                                                                                        |
|------------------|-------|-------------------------------------------------------------------------------------------------------------------|
| Surface measures | 5     | Mean sentence length, sentence-length standard deviation, mean word length, type-token ratio and stop-word ratio  |
| Function words   | 15    | Frequencies per 1,000 alphabetic words of a fixed function-word list                                              |
| Punctuation      | 6     | Counts of commas, full stops, colons, semicolons, question marks and exclamation marks per 1,000 alphabetic words |

The function-word list is *a, and, as, at, but, for, from, in, of, on, or, that, the, to, with*. Ratios reduce direct dependence on document size. They do not eliminate length effects entirely. In particular, type-token ratio tends to vary with text length. Punctuation frequencies count the corresponding spaCy punctuation tokens.

Features have different units, so the style vectors are standardised using reference-set means and population standard deviations:

$$z_{d,j} = \frac{x_{d,j} - \mu_{j,\mathcal{R}}}{\sigma_{j,\mathcal{R}}}.$$

The exclamation-mark feature is constant in the reference set and is removed. The resulting style representation has 25 dimensions. No test-set statistic is used in this transformation.

### 4 Models and comparison

The TF-IDF reference reuses the tokenizer, stemmer and vectoriser definitions from the Reuters notebook. The verified vocabulary contains 832 unigrams, 151 bigrams and 17 trigrams. Its pre-processing, including the stop-word consistency warning, is preserved.

The first style model uses the same nearest-neighbour rule as the TF-IDF model. Only the representation changes, including its associated preprocessing and scaling. The second style model calculates one mean vector for each author:

$$c_a = \frac{1}{|\mathcal{R}_a|} \sum_{d \in \mathcal{R}_a} z_d, \quad \hat{y}(z) = \arg \min_a \|z - c_a\|_2.$$

The centroid and Euclidean distance calculations use NumPy arrays. No hyperparameter search or test-driven feature selection is performed. All specified models are reported rather than choosing a winner and presenting its test score as a new validation result.

## 5 Results

The primary evaluation uses the 150 official test articles. Accuracy measures the proportion assigned correctly. Macro-F1 averages the F1 scores of the three authors with equal weight [3].

| Model                     | Mystery | Test correct | Test accuracy | Macro-F1 |
|---------------------------|---------|--------------|---------------|----------|
| TF-IDF, nearest neighbour | 3/3     | 137/150      | 91.3%         | 0.914    |
| Style, nearest neighbour  | 1/3     | 104/150      | 69.3%         | 0.686    |
| Style, author centroid    | 2/3     | 124/150      | 82.7%         | 0.828    |

All test classes contain 50 articles. Always predicting one fixed author would therefore achieve 33.3% accuracy. The style-centroid result is well above this reference, although it remains below the word-based nearest-neighbour result for this author trio.

The larger test changes the interpretation of the mystery demonstration. The TF-IDF method remains successful, but its three correct mystery predictions do not translate into perfect test performance. Conversely, the style nearest-neighbour model’s 1/3 mystery result understates its performance on the larger test set.

For the style-centroid model, the confusion matrix is:

| True author / predicted author | Kazer | Farrand | Commins |
|--------------------------------|-------|---------|---------|
| Kazer                          | 40    | 8       | 2       |
| Farrand                        | 11    | 39      | 0       |
| Commins                        | 3     | 2       | 45      |

Its recall is 80% for Kazer, 78% for Farrand and 90% for Commins. Confusion is concentrated between Kazer and Farrand. The TF-IDF model has a different error pattern: ten of its thirteen errors assign Commins articles to Farrand. The remaining errors are two Kazer articles assigned to Farrand and one Farrand article assigned to Kazer.

## 6 Interpretation of style and visualisation

The reference articles differ in several descriptive style measures:

| Author  | Mean sentence length | Mean word length | Commas / 1,000 words |
|---------|----------------------|------------------|----------------------|
| Kazer   | 22.91                | 4.96             | 42.25                |
| Farrand | 21.62                | 4.86             | 43.95                |
| Commins | 20.96                | 5.07             | 64.41                |

These are averages of document-level measurements across the 49 reference articles for each author. Commins’s reference articles contain more comma tokens per 1,000 words and somewhat longer words. Such differences help describe the data but do not establish which feature caused an individual prediction.

For matched PCA views, both representations are projected using an exact full-SVD solver fitted on the same reference articles. The two word-based components explain 6.23% and 5.35%, totalling 11.58%. The style components explain 11.22% and 9.71%, totalling 20.94%. The larger style percentage does not imply a better classifier because the representations have different dimensions and variance structures.

The generated comparison figure shows more overlap in the first two style components, while the TF-IDF view reflects strong lexical grouping. All three classifiers still operate in their full feature spaces. Two-dimensional distances in the plots are not the distances used for the reported predictions.

## 7 Qualitative error analysis

Three official-test cases illustrate disagreements between representations. Within each outcome category, the first sorted relative filename was chosen after evaluation. All three happen to be Commins articles. This is a post-hoc illustration, not a representative error sample or a tuning step. The full diagnostics are in `results/qualitative_cases.json`.

**Retail vocabulary across authors.** Commins’s `314014newsML.txt` discusses Dayton Hudson store closures. Its nearest TF-IDF reference is Farrand’s `173770newsML.txt`, on Burton’s profits. Shared features `store`, `retail`, `profit` and `divis` make the thematic overlap visible. TF-IDF predicts Farrand, whereas both style models correctly predict Commins. Her centroid distance is 3.020, compared with 3.660 to Kazer and 3.729 to Farrand.

**Related companies support a correct lexical decision.** Commins’s `334989newsML.txt` concerns Quaker’s Snapple business. The nearest reference, her `255733newsML.txt`, discusses a possible sale of the same business. `quaker` and `snappl` dominate their shared TF-IDF products. TF-IDF and the style nearest neighbour are correct, but the centroid predicts Kazer, at distance 4.079 versus 4.336 to Commins. Stop-word ratio and mean word length favour Kazer in the squared-distance comparison. A correct lexical label can therefore coexist with substantial topic overlap.

**All models agree incorrectly.** Commins’s `404426newsML.txt` describes CPC’s corn-refining business. All methods predict Farrand, whose nearest word-based reference, `220852newsML.txt`, discusses Tate and Lyle and corn processing. Shared features include `corn`, `sweeten` and `syrup`. The style-centroid distances are 19.839 to Farrand and 20.030 to Commins. Semicolon frequency contributes 9.661 to the squared-distance advantage of Farrand over Commins, the largest absolute feature contribution. A product enumeration uses this punctuation in the query. This exposes sensitivity to a rare standardised feature, not a general author-specific rule.

For each centroid comparison, feature contributions are computed as  $(z_j - c_{a,j})^2 - (z_j - c_{w,j})^2$ , where  $w$  is the predicted winner and  $a$  is the true author for an error, or the second-closest author for a correct prediction. Positive values favour the winner. These values explain the model’s distance calculation, not the causes of an author’s writing style.

## 8 Limitations and reproducibility

The experiment concerns one fixed author trio. The authors’ topics differ, and even function-word usage, sentence structure and punctuation can depend on subject matter or editorial practice. The scores do not establish that the models recover a stable personal style independently of topic. News stories may also be related across the supplied train and test partitions. The duplicate checks cover exact matches only.

The compact feature set was chosen for interpretability rather than optimised performance. spaCy’s sentence boundaries and token decisions have not been manually audited across the corpus. The centroid’s improvement over style-based nearest neighbours is compatible with a benefit from averaging, but this single comparison does not establish a general advantage of centroid models.

The repository entry point `run_analysis.py` records document identities, hashes, feature definitions, scaling parameters, predictions, confusion matrices and software versions. It uses `stylometry/corpus.py` to load and split texts, `stylometry/features.py` for both representations and `stylometry/models.py` for reference scaling and distances. `prepare_data.py` downloads and

selects the corpus. These files no longer depend on older local project folders.

A new Python 3.13.13 environment installed from the repository reproduced all document identities, style measurements, predictions and model metrics. Direct dependencies and the spaCy model are pinned. Five semantic tests and fresh-kernel execution of all three notebooks passed. The reports preserve original numerical values and their independent verification separately where small historical differences remain unexplained.

Together, the two contributions show that lexical and explicit style representations both provide useful attribution signals for these Reuters authors. The official test evaluation supports a narrower claim than *robust authorship recognition*: performance is promising for this fixed corpus and author selection, with unresolved topic and evaluation limitations.

## References

- [1] Liu, Z. (2006). *Reuter\_50\_50*. UCI Machine Learning Repository. [doi:10.24432/C5DS42](https://doi.org/10.24432/C5DS42).
- [2] spaCy. *Linguistic features*. [Online documentation](#).
- [3] scikit-learn. *Classification metrics*. [Online documentation](#).